Duty Assignment Before Grid Pricing in Electric-Bus Fleets: An Exact Projection, Support Bounds and Their Limits

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Research draft, revised 3 November 2026

Abstract

Can a bus operator commit to duties before observing grid prices? For two common charging windows, we prove that one price-independent matching per fleet size preserves the entire attainable load–cost set, including under common conversion losses and vehicle-to-grid exports. The grid then selects fleet size and charging, while alternative pairings of that size add no flexibility. In a charge-only extension with consecutive convex load thresholds, the physical-versus-convexified cost gap is strictly less than one bus-day fee, even with hard grid limits and local backup. We characterize exact support-failure intervals in a lossless subclass and distinguish them from cycling price updates. On public German peak-run data at 400 kWh, support fails in 7 of 2,664 sampled scenarios while undamped prices cycle in 397. On a German distribution-feeder study, physical minimum-bus duties match numerical conic lower estimates within one cent in 66 of 72 primary assignments; four are solver-infeasible and two retain wider bounds. Binding voltage limits require local supply at the largest far depot, but we find no additional capacity-charge saving from pairing. The reduction's boundary matters: in 64 general hourly scheduling instances, a zero-price duty commitment remains measurably suboptimal in 48 after a price-blind minimum-energy tie rule. These results identify when duty choice can precede pricing and where that simplification needs a separate certificate.

1 Introduction

A bus operator can often solve duty assignment before optimizing charging. The computational attraction is clear: retain the chosen services on each bus and adjust charging as electricity prices change. The unresolved question is what this commitment loses. A cheap duty may force energy into an expensive or grid-constrained window, and another assignment with exactly the same number of buses may offer a different charging set. A valid separation therefore needs a physical dominance argument, or a measured and bounded commitment penalty.

We give such an argument for two synchronized service batches. For each number of paired services, a matching minimizing mandatory early charging contains the load–cost set of every other matching at that count. It can be chosen before observing electricity prices or specifying the feeder cost. This is one representative per fleet size; prices and network feasibility may still determine which size to use. The distinction is essential outside the class, where hourly charging and transitions can change the ranking of committed duties.

Marginal-price support supplies a second diagnostic. Whole fleets create a nonconvex load–cost set: a charging profile can be attainable only with a more expensive fleet than a convexified planner assigns to it. This connects assignment to the established nonconvex-pricing problem in electricity markets [15, 17, 29, 31]. Fixed-population charging aggregates, however detailed [24–26], do not retain a choice of the population of vehicle duties.

This paper gives an exact account of that trade-off for a class that is simple enough to solve and recognizable enough to matter: buses serve a morning peak batch and an afternoon peak batch, may charge at the depot in a common midday window, and must be full again after an overnight window. Any bus may serve at most one morning and one afternoon run, and a declared compatibility graph lists the allowed pairings.

Contributions.

1. **Exact projection (Theorem 3).** For each matching cardinality m, all assignments with m pairs share every aggregate quantity except one: the total early energy their paired buses must receive. A matching minimizing that requirement contains the *entire* physical load–cost polygon of every other matching of the same size, at identical loads and fees. The complete projection is therefore a union of at most $\nu+1$ polygons with at most five vertices each, with common charging and discharging losses, a shared bidirectional connector and full overnight replenishment. Each point has a finite, serial, state-of-charge-feasible realization.

2. **Anatomy of the support gap (Proposition 6).** In the lossless case the gap $\Delta = D - \mathrm{CH}$ between physical and convexified planning values is positive only if the marginal-cost spread between the two charging windows reaches f/η, where f is the daily vehicle cost and η an increment between consecutive forced-charging thresholds. The gap is at most $\kappa\eta^2/8$, where κ bounds the slope of that spread. For an additive shift in the spread, Proposition 7 gives the exact failure intervals, including their endpoints' support status. Their total width depends only on the variation in marginal spread between the first and last forced-charging thresholds.

3. **Price dynamics (Theorem 10).** For the damped price-adjustment loop that posts the marginal cost of the last response, undamped prices are eventually periodic with period at most two and can cycle even when a supporting price exists. Sufficiently small constant steps converge at a strict vertex. At a face-interior optimum, prices become trapped in an $O(a)$ band. Convergence there requires an exact tie and eventual selection of the optimal physical endpoint. Diminishing steps recover the hull spread. With quadratic costs the step-weighted load and vehicle fee also converge to the hull solution.

4. **Where the reduction stops (Section 6).** A common transition energy per pair preserves the reduction. With pair-specific transition energy, replacement holds within each group having the same cardinality and total transition draw. For a complete, physically feasible lossless charge-only perfect layer we characterize exactly when a single matching suffices. A separate complete-graph construction gives a positive support gap with curvature in the midday load. With a third service batch, fleet feasibility alone is strongly NP-complete. On one public 37-service timetable outside the two-batch class, rational certificates establish a strictly positive gap.

5. **Calibrated evidence (Section 7).** Using the public timetables of 18 German operators [32, 33], we build morning and afternoon peak runs and compute when the obstruction can arise. With 400 kWh batteries, 15 of 18 operators admit a maximum-size pairing with zero mandatory midday charge, giving a flat cost frontier. Among the remaining three, the smallest critical marginal spread is 0.53 EUR/kWh at a 150 EUR/day vehicle cost. When failure occurs, the gap is below 0.06% of total daily cost. Undamped price iteration nevertheless cycles in 15% of scenarios, nearly all of which have no gap. A German feeder comparison resolves numerical duty-first cost brackets within one cent in 66/72 primary assignments (Section 8). The separate 64-instance hourly test in Section 6 records the limits of zero-price commitment, including an energy-aware tie control.

The support criterion itself, matching integrality and convex-hull pricing are classical tools, and we use them as such. The new content is the exact physical description of the assignment-dependent load–cost set, together with the consequences it makes explicit for support and price iteration. The model is deliberately stylized: it abstracts from travel between services, staggered availability, battery degradation and switching time. Section 6 shows which of these abstractions are essential.

Organization. Section 2 positions the work. Section 3 defines complete plans and support. Section 4 proves the projection theorem, Section 5 the price-formation results, and Section 6 the limits of the reduction. Section 7 reports the tariff sensitivity study, and Section 9 discusses implications. The appendices give the proofs and distinguish reproduction from the committed peak-run inputs from verification of the separately held public-case certificates (Appendix F).

2 Related work

Electric-bus scheduling with charging. Exact and heuristic methods for the electric vehicle scheduling problem combine service covering with charging decisions under station capacity [10], scale through column generation [30], or integrate flow and set-partitioning models [7]. Wu et al. [35] add power-grid characteristics and endogenous vehicle-to-trip assignment with overnight charging. Nath et al. [27, Sec. 4.2] jointly choose charging locations, trip assignments and charging schedules under energy and contracted-power charges. At each fleet size, Dabčević et al. [9, Sec. 4.2] vary objective weights to trace the trade-off between deadheading and charging cost at given tariffs. Our question concerns the joint set of loads and fleet costs over *all* assignments, from which one can test support across tariffs.

Buses, tariffs and markets. Wu et al. [36] embed bus operations in a bi-level model with distribution locational marginal prices. Their upper level is a tactical frequency-setting problem with terminal opportunity charging; the lower level clears a DC power-flow market, and the bi-level problem is solved through its complementarity reformulation. Vehicle-to-service assignment is not modeled, so the assignment nonconvexity studied here does not arise. Zoltowska and Lin [38] aggregate bus charging into day-ahead auction bids; Blatiak et al. [6] optimize commercial-fleet trip timing and charging against energy and balancing prices; and Zhou et al. [37] formulate a multi-depot mixed-integer charging and discharging schedule for electric buses under vehicle-to-grid tariffs and solve it with an adaptive large-neighbourhood search (our comparison with the last is limited to its published abstract). All of these studies address operational scheduling or market participation. Our comparison concerns the proof that one assignment realizes every other assignment's attainable loads at the same fleet size.

Aggregate flexibility. For a fixed EV population, aggregate flexibility is a Minkowski sum of individual sets [3, 24]. Mukhi et al. [25, Thms. 3 and 7] characterize that sum and disaggregate its schedules for generalized polymatroids. Müller et al. [26, Eqs. (15)–(17)] also compute the least cost of disaggregation for zonotopic resources with structured costs. Arvis et al. [5] embed aggregate EV flexibility in unit commitment, and Al Taha and Bitar [2] give a state-of-charge reformulation for a single lossy storage device. In our setting a matching determines which duties enter the sum. At a fixed number of pairs the assignments have the same fleet cost and aggregate battery bounds. Only their mandatory early purchases differ, so one matching realizes the union of all their load polygons (Theorem 3 and Remark 4).

Nonconvex pricing. Uplift, convex-hull pricing and their computation are established [4, 15, 17, 29, 31]. For linear market participants with a complete linear description, Hümbs et al. [18, Thm. 1] characterize support by the existence of an integral optimum of the relaxed planner's problem. Proposition 2 uses the corresponding convex-hull criterion for a complete fleet and a convex supplier. Our gap bound uses the distance between adjacent fleet choices. Shapley–Folkman bounds instead relate the effect of individual nonconvexities to market aggregation [16, 34].

Price adjustment. Ma et al. [22, Alg. 1, Eq. (32), Cor. 4.1] update posted prices toward the marginal generation cost of the latest EV response and prove convergence under a contraction condition. We study this update with a discontinuous response that selects physical layer endpoints. For an affine marginal-cost spread g and an initial price $\pi_0 = g(q_0)$ at a feasible early load q_0, the update is the load component of Frank–Wolfe on the hull [19, Alg. 1]. Its linear oracle minimizes $c + g(q_k)E$, and $q_k = g^{-1}(\pi_k)$ satisfies $q_{k+1} = (1 - a_k)q_k + a_k E_k$. Ergodic recovery in convex dual methods is also classical [20, 28]. Theorem 10 identifies which fleet responses recur and how the face they expose determines support. Within its constant-step trapping band, the two endpoint branches, away from exact ties, rescale to the contracted rotation studied by Laurent and Nogueira [21].

3 Complete fleets and the meaning of support

Fix mandatory services, movement possibilities, a charging horizon divided into market intervals $t = 1, \dots, T$, and physical equipment. A *complete plan* $x \in \mathcal{X}$ covers every mandatory service exactly once, assigns services to used buses, routes them, and schedules charging subject to battery and shared-resource limits. Let $e(x) \in \mathbb{R}^T$ be the fleet's net grid energy by interval (negative for exports) and $c(x)$ its intrinsic operating cost, here the vehicle cost. The projected image $S = \{(e(x), c(x)) : x \in \mathcal{X}\}$ is assumed nonempty and compact.

A supplier serves the fleet's incremental demand at a proper closed convex cost $F : \mathbb{R}^T \to \mathbb{R} \cup \{+\infty\}$, finite on all attainable loads. The *physical* and *convexified* planning values and the *gap* are

$$D = \min_{x \in \mathcal{X}} \{c(x) + F(e(x))\}, \qquad \mathrm{CH} = \min_{(e,c) \in \operatorname{conv} S} \{c + F(e)\}, \qquad \Delta = D - \mathrm{CH} \geq 0. \tag{1}$$

The hull mixes complete plans. Its optimum is a schedule mixture, not one executable day. At a posted price vector $p \in \mathbb{R}^T$ the fleet's response value and the regret of plan x are

$$V(p) = \min_{y \in \mathcal{X}} \{c(y) + p^\top e(y)\}, \qquad r(x; p) = c(x) + p^\top e(x) - V(p). \tag{2}$$

Definition 1 (Own-load support)**.** A physical plan x is *supported* if some marginal price $p \in \partial F(e(x))$ gives $r(x; p) = 0$.

The operator in (2) controls the complete fleet and takes the posted price as fixed. This is the decentralization test behind marginal-cost tariffs: a price computed from the operator's own load must not give it a reason to deviate. A strategic operator anticipating the price change would solve a different problem, which we do not study.

Proposition 2 (Classical support criterion)**.** *For every physical x and $p \in \partial F(e(x))$, $r(x; p) \geq c(x) + F(e(x)) - \mathrm{CH} \geq \Delta$. If $\operatorname{relint} \operatorname{conv} e(\mathcal{X}) \cap \operatorname{relint} \operatorname{dom} F \neq \emptyset$, then some physical plan is supported if and only if $\Delta = 0$, and in that case every physical optimum is supported by every maximizer of $V(p) - F^*(p)$.*

This specializes standard arguments [15, 17, 29]; the proof is in Appendix B. Two consequences are used below. A positive gap gives every physical plan a regret of at least Δ at its own marginal prices. Conversely, at a maximizer p^* of $V(p)-F^*(p)$ (a convex-hull price), the identity in the proof shows that a physical optimum's regret is at most Δ, so a payment conditional on implementing that particular schedule can remove the operator's incentive to deviate at the hull price. A strict gap still excludes own-load support as defined above.

4 An exact assignment-dependent projection

The two-batch class. There are $n_A \geq 1$ morning (A) services during $[0,1]$ and $n_B \geq 1$ afternoon (B) services during $[2,3]$, with battery draws $0 < a_i, b_j \leq B$ for a common usable capacity $B > 0$. Same-batch services overlap, so each bus serves at most one A and one B service. A bipartite graph Γ lists the allowed A–B pairings; every service may also be served by its own bus. A partial matching $M \subseteq \Gamma$ with $|M| = m$ uses $N_m = n_A + n_B - m$ buses, each at cost $f > 0$; let ν be the maximum matching size. Transitions take no time or energy (Section 6 relaxes this). All buses start full, are at the depot throughout the *early* window $[1,2]$, and must be full again after the *terminal* window $[3,4]$. One shared early connector moves at most $K \geq 0$ grid-side energy in either direction; one terminal connector delivers at most $H \geq 0$, charging only. Every bus accepts the full connector power throughout each corresponding unit window. Service draw is nonnegative throughout a run, and there is no auxiliary draw outside the specified tasks. Charging and discharging efficiencies are common, $\alpha, \delta \in (0,1]$. Early actions may change direction finitely often, are sequential on the connector, and incur no switching delay or cycling cost.

Let C and X be the integrated early purchases and exports. The connector's gross use is $C + X \leq K$ and the net early load is $E = C - X$. Write

$$S_0 = \sum_i a_i + \sum_j b_j, \qquad P = \sum_i a_i, \qquad Q = \alpha C - X/\delta.$$

Q is the fleet's net battery gain in the early window. Full replenishment then requires the terminal purchase $L = (S_0 - Q)/\alpha$. The cost F is a function of the two window loads (E, L).

Theorem 3 (Complete common-window projection)**.** *For $m = 0, \ldots, \nu$ let*

$$r_m = \min_{\substack{M \text{ matching in } \Gamma \\ |M|=m}} \sum_{(i,j)\in M} (a_i + b_j - B)_+,$$

$$W_m = \Big\{(C,X):\ C \geq r_m/\alpha,\ X \geq 0,\ C + X \leq K,\ \max\{S_0 - N_m B,\ S_0 - \alpha H\} \leq \alpha C - X/\delta \leq P\Big\}.$$

The set of attainable (early load, terminal load, vehicle cost) triples is exactly

$$\mathcal{P} = \bigcup_{m=0}^{\nu} \Big\{(C - X,\ (S_0 - \alpha C + X/\delta)/\alpha,\ fN_m) : (C,X) \in W_m\Big\}.$$

Every point has a realization with finitely many serial actions. Each nonempty W_m is a polygon with at most five vertices, so conv $\mathcal{P}$ *is generated by at most $5(\nu+1)$ points. If exports are prohibited, the same holds with $X = 0$.*

Why one matching statistic suffices. Fix a matching. A bus's net early battery gain x_v lies in an interval: $[a_i + b_j - B,\, a_i]$ for a paired bus, $[a_i - B,\, a_i]$ for an A-only bus and $[b_j - B,\, 0]$ for a B-only bus. The lower endpoints sum to $S_0 - N_m B$ and the upper endpoints to P, whatever the matching. Only the sum R of the *positive* lower endpoints, the energy that paired buses must receive before their afternoon service, depends on which services are paired. At any admissible total Q, the intervals can be filled so that positive gains total $\max\{R, Q\}$, giving one-direction purchases $\max\{R, Q\}/\alpha$ and exports $\delta(\max\{R, Q\} - Q)$. Any additional purchase $z \geq 0$ with matching exports $\alpha\delta z$ is realized by battery-neutral charge–discharge cycles on one bus, in chunks of depth at most $B/2$. Serializing all actions uses exactly $C + X$ of connector capacity. At fixed m, lowering R only relaxes the constraint $C \geq R/\alpha$, so a minimum-requirement matching contains the polygon of every other matching of that size. This is set equality, not just economic dominance; the complete proof is in Appendix A.

Remark 4 (What a net-interval aggregate loses). Two fleets whose battery-gain intervals sum to the same net interval can have different mandatory purchases: $[1, 2]$ and $[-1, 0]$ sum to $[0, 2]$, as do $[0, 1]$ and $[0, 1]$, but the first pair forces at least one unit of purchase. With losses and a gross connector budget this difference changes the attainable (E, L) set. The statistic r_m is exactly the information about the assignment that the aggregate must keep.

The lossless charge-only case. With $\alpha = \delta = 1$ and no exports, each layer reduces to an interval of early loads. Let $\mathrm{Lo} = \max\{0, S_0 - H\}$, $U = \min\{P, K\}$ and $\ell_m = \max\{\mathrm{Lo}, r_m\}$. Then

$$\mathcal{P} = \bigcup_{m:\ \ell_m \leq U} \{(E,\, S_0 - E,\, fN_m) :\ \ell_m \leq E \leq U\}, \tag{3}$$

the model is feasible exactly when $\mathrm{Lo} \leq U$, and r_m is nondecreasing and discretely convex in m (Appendix A). Let m_0 be the largest m with $r_m \leq \mathrm{Lo}$ and $m_{\max}$ the largest m with $\ell_m \leq U$. The lower boundary $\bar{c}(E)$ of the hull's operating cost is piecewise linear. Call Lo and r_m for $m_0 < m \leq m_{\max}$ the *breakpoints*. The boundary starts at (Lo, fN_{m_0}), interpolates linearly between consecutive breakpoints (ℓ_{m-1}, fN_{m-1}) and (ℓ_m, fN_m), and is flat at $fN_{m_{\max}}$ from the last breakpoint to U (the *flat face*). Consecutive breakpoints are one cardinality apart, so each sloping segment has slope $-f/\eta_m$ with increment $\eta_m = \ell_m - \ell_{m-1} > 0$.

Corollary 5 (Support classification)**.** *Let $G(E) = F(E, S_0 - E)$ be continuous and strictly convex on $[\mathrm{Lo}, U]$ and let E^* minimize $\bar{c} + G$. Every early load in $[\mathrm{Lo}, U]$ is attainable. A physical plan attains the hull cost $\bar{c}(E^*)$ at that load if and only if E^* is a breakpoint or belongs to $[\ell_{m_{\max}}, U]$. In those cases $\Delta = 0$. The gap is positive exactly when E^* lies strictly between consecutive breakpoints. Under the relative-interior condition of Proposition 2, the zero-gap cases are exactly those admitting own-load support.*

For a complete graph, r_m has a closed form: sort both batches increasingly and pair the m smallest morning draws with the m smallest afternoon draws in opposite order, $r_m = \sum_{i=1}^{m} (a_{(i)} + b_{(m+1-i)} - B)_+$ (a classical exchange argument, Appendix A). For a general graph, r_m is a minimum-cost matching of cardinality m.

A worked example. Take morning draws $(5, 15)$, afternoon draws $(14, 15)$, capacity $B = 20$, $K = 15$, $H = 49$, $f = 7$, and the complete graph. Then $(r_0, r_1, r_2) = (0, 0, 9)$: the two-bus crossed pairing needs 9 kWh of midday charge, while pairing by index would need 10. With $F(E, L) = 4E + (E^2 + L^2)/20$, the physical optimum is the two-bus plan at $E = 9$, with $D = 2681/20$. The hull optimum lies at $E^* = 151/18$ on the segment joining the three-bus plan at $E = 0$ to the two-bus

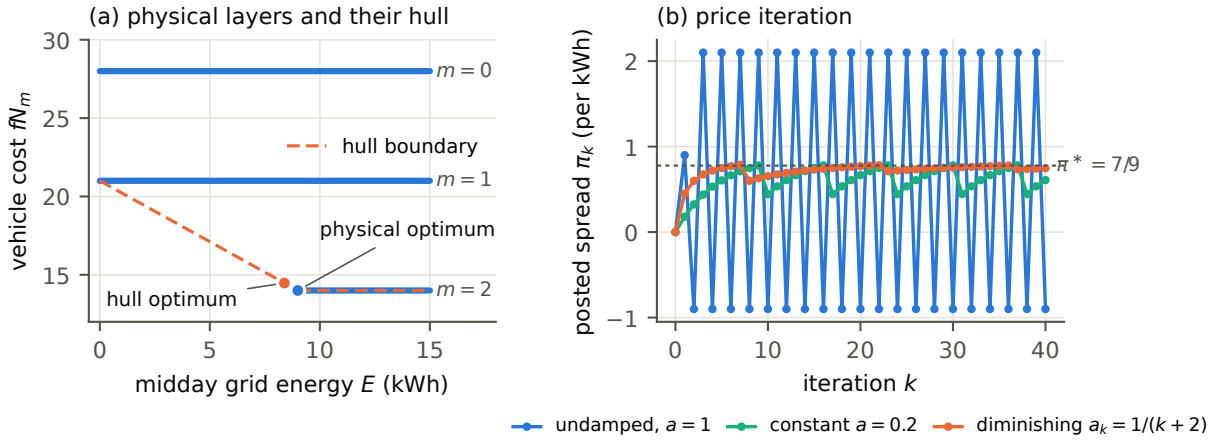


Figure 1: The worked example. (a) Physical layers (solid) and the lower hull boundary (dashed); the hull optimum lies strictly inside the sloping face, so $\Delta > 0$. (b) Price iteration on the scalar spread $\pi = p_E - p_L$: undamped prices enter a 2-cycle that alternates between loads 0 and 15. A constant step $a = 0.2$ keeps spreads in a band around the convex-hull spread $\pi^* = 7/9$ and repeatedly visits loads 0 and 9. Diminishing steps converge to π^*.

plan at $E = 9$, with CH $= 434201/3240$ and $\Delta = 121/3240$ (Figure 1a). At its own marginal prices the physical optimum has regret $11/10$, about thirty times Δ: the support failure costs the operator far more in bill terms than the planning gap costs the system.

5 Price formation on the exact class

We study the lossless class (3). With bidirectional exchange the layers are $[t_m, U]$ with $t_m = \max\{S_0 - H,\ S_0 - N_m B,\ 2r_m - K\}$. The results below apply with $\mathrm{Lo} = t_0$, $\ell_m = t_m$, $m_0 = \max\{m : t_m = t_0\}$ and $m_{\max} = \max\{m : t_m \le U\}$ (Appendix A).

On the physical set, the terminal load is $S_0 - E$. A posted price $p = (p_E, p_L)$ therefore charges $p_L S_0 + \pi E$ with $\pi = p_E - p_L$, and the operator's response depends only on the *spread* π. Assume $\mathrm{Lo} < U$, and let F be convex and differentiable on a neighbourhood of the attainable loads. Let $g(E) = \partial_E F - \partial_L F$ at $(E, S_0 - E)$. This is the marginal-cost spread at early load E, and it equals $G'(E)$. Assume g is continuous and strictly increasing on $[\mathrm{Lo}, U]$, let E^* be the unique hull optimum, and let $\pi^* = g(E^*)$, the *convex-hull spread*. We study a fleet solver that returns a layer endpoint. Write $R(\pi)$ for the loads of all physical layer endpoints minimizing $fN_m + \pi E$. This includes an endpoint at a collinear breakpoint when several cardinalities tie. At $\pi = 0$, interior loads of the cheapest layer also minimize the bill and lie outside this selection rule.

Anatomy of the gap.

Proposition 6 (Gap conditions)**.** *Let* $\eta_{\max} = \max_{m_0 < m \le m_{\max}} \eta_m$ *(zero if there is no sloping face).*

(i) If $\Delta > 0$*, then* E^* *lies strictly between consecutive breakpoints* $u < u'$ *with* $\pi^* = f/(u' - u)$*. In particular,* $\eta_{\max} > 0$ *and* $\max_{[\mathrm{Lo},U]} g > f/\eta_{\max}$*.*

(ii) If g is κ-Lipschitz, then $\Delta \le \kappa\eta_{\max}^2/8$. When $\Delta > 0$, the sharper bound is $\kappa(u'-u)^2/8$ for the interval in (i).

For an optimum between consecutive breakpoints, the marginal spread times the forced early energy equals the vehicle cost saved by one more pair. Varying the spread intercept identifies the supply curves for which a mixture improves on every physical schedule.

Proposition 7 (Exact intercept bands)**.** *Fix the physical layers and $f > 0$. Let $G'_s(E) = s + h(E)$ for $s \in \mathbb{R}$, where h is continuous and strictly increasing. The intercepts producing a positive gap are precisely*

$$\{s : \Delta(s) > 0\} = \bigcup_{m=m_0+1}^{m_{\max}} \left(\frac{f}{\eta_m} - h(\ell_m),\ \frac{f}{\eta_m} - h(\ell_{m-1}) \right). \tag{4}$$

These open intervals are disjoint. Their total length on the real line is $h(\ell_{m_{\max}}) - h(\mathrm{Lo})$, independent of $f > 0$. If there are no sloping segments the union is empty and its length is zero.

For quadratic window costs with $h(E) = \beta(2E - S_0)$ and $\beta > 0$, the total length is $2\beta(\ell_{m_{\max}} - \mathrm{Lo})$. The fee sets the location of each interval, so the length within a given tariff range and the number of hits on a finite grid can depend on f.

Proposition 8 (One-fee rounding bound)**.** *In a charge-only two-window projection, suppose consecutive pair counts $m = 0, \dots, \nu$ have early-energy intervals $[\ell_m, U]$ with a common upper endpoint, nondecreasing discrete-convex ℓ_m, and fee $f(N-m)$, where $f > 0$. Let $G(E)$ be any common grid and supply cost, possibly taking value $+\infty$ to impose hard feasibility. If the convexified optimum is finite and attained at mean count t and energy E, then*

$$0 \le D - \mathrm{CH} \le f(t - \lfloor t \rfloor) < f.$$

Proof. Let $\ell(t)$ interpolate consecutive lower endpoints. Discrete convexity makes $\ell(t)$ convex. Jensen's inequality and mixing the neighboring layers show that the hull at mean count t has exactly the energy interval $[\ell(t), U]$. Its fee is $f(N-t)$. The integer count $\lfloor t \rfloor$ admits the same E, because $\ell_{\lfloor t \rfloor} \le \ell(t) \le E \le U$. Keep its entire grid dispatch and any backup supply. Only the fee changes, by $f(t - \lfloor t \rfloor)$. A physical optimum can do no worse. □

The constant is sharp. Two services each draw one unit, the usable battery holds one, and pairing forces one unit of early charge. Two singleton buses admit $[0, 1]$. With free supply below an early cap $1 - \varepsilon$ and infeasibility above, subtract the common one-bus fee. The physical cost is f and the hull cost $f\varepsilon$, approaching a gap of f. The bound applies to the entire schedule, irrespective of the number of binding network constraints. It does not extend by counting binding windows in an arbitrary multidimensional scheduling problem.

A cap between forced levels is insufficient for a positive gap: the optimum may occur elsewhere or at a supported tie. Price ambiguity is also unnecessary. With fee one on $[0, 1]$, fee zero at $E = 1$, and smooth cost $G(E) = E^2$, physical cost is one and hull cost is $3/4$, at the unique marginal price one. Support failure and an update orbit need separate tests.

Price iteration. A natural way to find a supporting price is to post a price, observe the fleet's response, and move the price toward the marginal cost of that response:

$$\pi_{k+1} = (1 - a_k)\pi_k + a_k\, g(E_k), \qquad E_k \in R(\pi_k), \quad a_k \in (0, 1]. \tag{5}$$

Applied componentwise to (p_E, p_L), the damped update $p_{k+1} = (1 - a_k)p_k + a_k \nabla F(e_k)$ induces (5) on the spread. At a fixed spread $\pi = g(E)$ the selected plan is supported by its own marginal-price

vector. We call a corner or domain endpoint of the lower frontier a *strict vertex*. A maximal straight segment of positive length is a *face*. Let $\mathcal{T}$ consist of the negatives of the slopes of these faces, the spreads at which their endpoints tie.

Lemma 9 (Sign property)**.** *For every π and every $E \in R(\pi)$, $(\pi-\pi^*)(g(E)-\pi^*) \le 0$. Consequently, for every step $a \in (0,1]$, $|\pi_{k+1}-\pi^*| \le \max\{(1-a)|\pi_k-\pi^*|, aM\}$, where $M = \max_{E\in[\mathrm{Lo},U]} |g(E)-\pi^*|$.*

Theorem 10 (Price iteration on the two-batch class)**.** *Under the standing assumptions of this section, the following statements hold.*

(a) Fixed points. *Some vertex selection has a fixed point if and only if E^* is a breakpoint or $E^* = U$. In particular there is no fixed point when $\Delta > 0$, nor when E^* lies inside the flat face, although $\Delta = 0$ in the latter case.*

(b) Undamped iteration *($a_k \equiv 1$, a deterministic selection). Every orbit is eventually periodic with period one or two. If E^* lies in the relative interior of a maximal face $[w, w']$ and $\pi_0 \neq \pi^*$, the period is two and the cycle's loads satisfy $E_{\mathrm{lo}} \le w < w' \le E_{\mathrm{hi}}$.*

(c) Constant step *$a \in (0,1)$. Every orbit enters the interval $|\pi - \pi^*| \le aM$ after finitely many steps and remains there. If E^* is a strict vertex and $aM < \mathrm{dist}(\pi^*, \mathcal{T}\setminus\{\pi^*\})$ (distance to the empty set is infinite), then $E_k = E^*$ eventually and $\pi_k \to \pi^*$ geometrically. If E^* lies inside a face $[w, w']$, convergence requires an exact hit on π^* followed by eventual selection of E^* at every step. This is possible for at most countably many initial spreads and requires E^* to be a breakpoint. Under the same step bound, all sufficiently late responses lie on $[w, w']$ and*

$$\pi_k \in [\pi^* - a\,d_-,\ \pi^* + a\,d_+], \qquad d_- = \pi^* - g(w), \quad d_+ = g(w') - \pi^*.$$

Within this band the response is w above π^* and w' below π^*. Both endpoints recur infinitely often if the orbit eventually avoids π^* or if the tie rule at π^* always chooses an endpoint of the maximal face.

(d) Diminishing steps *($a_k \to 0$, $\sum_k a_k = \infty$). Then $\pi_k \to \pi^*$, and the step-weighted averages of $g(E_k)$ converge to π^*. If g is affine, the step-weighted averages of load and vehicle cost converge to E^* and $\bar{c}(E^*)$, respectively.*

Lemma 9 follows from monotonicity of the hull frontier's subdifferential (Appendix C). Figure 1b illustrates all three regimes on the worked example. The step-size conditions in (c) matter. In the worked example with the steeper supply $F(E,L) = 16E + (E^2+L^2)/4$, the hull optimum is the supported vertex $E^* = 9$, yet undamped iteration from $\pi_0 = 3$ alternates between loads 0 and 15 forever, while the step $a = 1/10$ converges. Part (b) also assumes a selection that depends only on the current price; history-dependent tie rules are outside its scope.

Ties and support. An optimum strictly between consecutive breakpoints has a positive gap. A breakpoint in the interior of a longer sloping face has zero gap and can be selected at the exact tie. An optimum inside the flat face also has zero gap. The cheapest fleet can deliver it, although the endpoint rule selects only the ends of that layer. These cases can all exhibit persistent spread fluctuations. The empirical frequencies in Section 7 refer to the stated tie rule and initial spread.

An exact tie can change which loads recur. Take three services in each batch, all draws and B equal to one, a complete graph, $K = 3$, $H = 6$ and $f = 1$. Here $\ell_m = m$ and $\bar{c}(E) = 6 - E$ on $[0,3]$. With $F(E,L) = E^2/2 + (4/3)L$, the optimal spread $\pi^* = 1$ occurs at $E^* = 7/3$. At $\pi = 1$ let

the solver choose the physical endpoint $E = 1$. Starting there with $a = 1/2$ gives the spread cycle $1, 1/3, 1$ and the repeating loads $1, 3$. The maximal face has left endpoint 0, which this orbit never visits.

Averaging. With an affine spread and diminishing steps, part (d) recovers the hull load and vehicle cost using weights $a_k / \sum_{j<n} a_j$. Convexity gives $G(\sum_k \lambda_k E_k) \le \sum_k \lambda_k G(E_k)$ for any mixture weights λ_k. Every executed day costs at least D, even when the weighted load and vehicle cost converge to a hull point of cost $\mathrm{CH} < D$. A constant step gives the band in (c). Its daily loads can have a mean different from E^*.

6 Where the reduction stops

The reduction rests on one fact: at fixed cardinality, total energy and every aggregate bound except the mandatory purchase are independent of the matching. We now show which changes preserve that fact and which destroy it.

Common transition energy preserves it. If every matched pair performs the same compulsory task of energy $d \ge 0$ during a noncharging interval between the early window and the B batch, then total consumption is $S_0 + md$, the paired interval becomes $[a_i + b_j + d - B, a_i]$, and edges with $b_j + d > B$ are removed. Theorem 3 holds with these substitutions (Appendix D).

Proposition 11 (Matching replacement with transition energy)**.** *In the same noncharging interval, let edge (i, j) require a compulsory draw $d_{ij} \ge 0$. Discard edges with $b_j + d_{ij} > B$. Group matchings by cardinality m and transition total $z = \sum_{(i,j)\in M} d_{ij}$. In each nonempty group a matching minimizing $R(M) = \sum_{(i,j)\in M} (a_i + b_j + d_{ij} - B)_+$ realizes the whole group's set of (E, L, fN_m). Its polygon is given by Theorem 3 with $S_0 + z$ in place of S_0 and this minimum in place of r_m, including losses and exports. Thus an invariant transition total suffices for one matching to represent a cardinality layer. In particular, if $d_{ij} = u_i + v_j$ on the allowed edges, every perfect matching of two equal batches has total $\sum_i u_i + \sum_j v_j$.*

For the lossless charge-only perfect layer of a complete $n \times n$ graph with $n \ge 2$, suppose every perfect matching admits a physical plan. One perfect matching realizes every attainable triple of that layer if and only if all perfect matchings have the same transition total. This is equivalent to $d_{ij} + d_{i'j'} = d_{ij'} + d_{i'j}$ on every 2×2 submatrix, or equivalently to $d_{ij} = u_i + v_j$ for row and column potentials.

When transition totals differ. Different totals can require different representative matchings.

Proposition 12 (Support failure on a complete assignment graph)**.** *For every $k \ge 1$ there is a charge-only two-batch instance with $2k$ services per batch, the complete assignment graph, pair-dependent compulsory transition energies, and a fleet cap of $2k$, in which every perfect matching is physically feasible and, for the strongly convex-in-E supply $F(E, L) = 3E + L + (E - 3)^2/(2K)$ and any $f \ge 0$,*

$$D = \mathrm{CH} + \frac{9}{2K} > \mathrm{CH}.$$

The construction and proof are in Appendix D. Every assignment alternative is admitted, so the gap is not an artifact of a sparse compatibility graph or of infeasible pairings. The gap shrinks as K grows; the statement is an existence result, not a prevalence claim.

A third batch makes feasibility hard.

Proposition 13 (Three batches)**.** *With three successive batches of n simultaneous services, complete zero-energy compatibility between consecutive batches, no charging before the last batch, fleet cap n and an adequate terminal connector, deciding whether a feasible fleet exists is strongly NP-complete.*

The proof reduces from numerical three-dimensional matching [14, problem SP16] (Appendix D). It rules out a general polynomial-time extension of the two-batch algorithm to this three-batch feasibility problem unless P equals NP.

Positive setup time. The construction in Theorem 3 uses arbitrarily many short neutral cycles and zero switching time. A positive setup time per action breaks exactness: it can make a layer infeasible while neighbouring layers stay feasible, and the number of buses needing early charge then matters separately from r_m. The companion setup-time proof in the repository bounds the change in the gap when every relevant layer has a feasible point with sufficient unused connector time. Whether actual depot windows provide enough slack for this bound to be informative remains open.

Beyond the class: one certified public timetable. The 37 services in the Hildenbrand timetable of Sistig et al. [32] have staggered start times. We use its directed deadhead matrix and model charging at one depot with a shared 360 kW connector, 400 kWh batteries, a vehicle cost of 100 and a synthetic quadratic hourly supply cost. A mixture of two feasible plans bounds CH from above. A rational branch proof and separate fleet-count bounds cover every physical plan. Two energy models give the same outward bounds,

$$\mathrm{CH} \le 441.142238 < 441.532912 \le D \le 441.532913, \qquad D - \mathrm{CH} \ge 0.390674.$$

The certified separation is at least 0.08848% of D. The hull optimum and the exact gap remain unknown. By Proposition 2, no physical plan is supported at its own marginal prices. The first certificate uses modeled consumption stored on a finite movement graph. The second uses exact additive consumption and extends to finite walks over the specified location pairs under the travel and depot-access assumptions in Appendix E.

A duty-first boundary test on hourly schedules. We also use all 64 previously certified synthetic development instances (16 seeds, 8 or 12 trips, and two affine supply slopes). These have hourly depot charging, travel between services, 60 kWh batteries and terminal reserve rather than full replenishment. Costs are in SEK. Choose operating-cost-minimum duties at zero electricity price, retain their ordered trip partition, and reoptimize charging and direct/depot transitions against the original system objective. Adaptive tangent MILPs and the original joint certificates give a regret interval for every cell. All 64 fixed-duty problems meet a 0.01 SEK optimality tolerance. The arbitrary zero-price tie gives a regret lower endpoint above 0.01 in 53 cases, mean regret interval $[12.192, 12.207]$ SEK, and maximum upper endpoint 54.094 SEK.

To test the role of that tie, a separate prespecified sensitivity chooses minimum charging energy among the zero-price operating-cost optima, before observing the time-dependent prices. It still loses more than 0.01 SEK in 48 of 64 cells. The mean regret interval is $[7.753, 7.769]$ SEK and the largest upper endpoint 46.126 SEK; the largest total-cost percentage upper bound is 2.043%. No interval contradicts the original joint certificate. This marks a boundary of these duty-first rules. It does not rule out a better price-independent rule or establish a population claim. The historical

joint bounds were reused, with the original instance and market sources; neither the German feeder nor its tariffs were attached to these synthetic cases.

7 Tariff sensitivity on public peak-run data

The theory says when support can fail and how large the gap can be. We now ask whether those conditions arise for run energies derived from public timetables.

Data and instance construction. The dataset of Sistig et al. [32] (CC BY 4.0) contains timetables, stops and directed deadhead matrices for German bus operators. For each of the 20 operators in the archive we choose the day type with the most service records after the speed screen, then retain trips whose endpoints appear in the deadhead matrix. Trips departing in the morning peak $[05{:}30, 09{:}30)$ form the A batch and trips departing in the afternoon peak $[14{:}30, 18{:}30)$ form the B batch. Within each batch one maximum matching on the reachability graph partitions trips into a minimum number of vehicle runs. Ties between minimum-run covers may give different run energies. We condition on the stored cover. Each run's energy uses the source study's vehicle parameters: 1.2 kWh/km in service, 0.96 kWh/km deadheading, and 16 kWh/h of auxiliary load away from the depot, including pull-out and pull-in using separately chosen minimum-energy depot legs [33]. We exclude 108 trip records whose implied speed exceeds the 90 km/h screen and the two operators with fewer than five runs per batch. This leaves 18 operators with 6–413 morning runs and 6–298 afternoon runs. The 3913 run energies have median 154 kWh and interdecile range 110–204 kWh. We retain these modeled run energies, impose common charging windows and allow complete cross-batch pairing. The resulting dispatches describe peak-run abstractions. Full-day service, crews, consistent depot assignments and alternative within-peak chainings require separate validation.

Economics. We treat economics as a sensitivity grid rather than a calibration. The daily vehicle cost is $f = 150$ EUR in the sweeps; critical spreads scale linearly in f. The supply cost in each window is quadratic, $\bar{p}_w E + \beta E^2/2$, so the marginal-cost spread is $g(E) = s_0 + \beta(2E - S_0)$, which is Lipschitz with $\kappa = 2\beta$. The zero-load spread $s_0 = \bar{p}_E - \bar{p}_L$ ranges over 37 values from -0.2 to 1.6 EUR/kWh. The price impact is $\beta = \rho\,\bar{p}/S_0$ with $\bar{p} = 0.15$ EUR/kWh and $\rho \in \{0.5, 1, 2, 4\}$: drawing the fleet's whole daily energy in one window would raise that window's marginal cost by $\rho\bar{p}$. We report costs relative to the total daily cost, vehicles plus energy, with the overnight zero-load price set to $\bar{p}$. The midday budget K is either ample ($K = P$) or half that, and the overnight budget is ample. The model has no in-service charging, so a run needing more than the usable battery B cannot be served. We exclude such operator–battery combinations rather than trimming their runs (57 runs exceed 250 kWh, 13 exceed 300 kWh, 8 exceed 350 kWh, none exceed 400 kWh).

Which operators admit a zero-requirement matching? Proposition 6(i) turns the data into a necessary condition: a gap requires a marginal spread of at least $f/\eta_{\max}$, which depends only on run energies and battery size. Figure 2 shows it per operator. With $B = 400$ kWh, 15 of 18 operators admit a maximum-size matching with zero mandatory midday charge. Removing pairs gives $r_m = 0$ at every smaller size. The cheapest fleet therefore attains the entire feasible load interval. Only six of the 18 satisfy the stronger condition that *every individual edge* has zero requirement. For the three operators with a positive matching requirement the critical spread is 0.53 EUR/kWh or more, and above 35 EUR/kWh for two of them. Smaller batteries make forced charging real. At $B = 350$ kWh, 5 of 17 eligible operators admit a gap, from 0.81 EUR/kWh (median

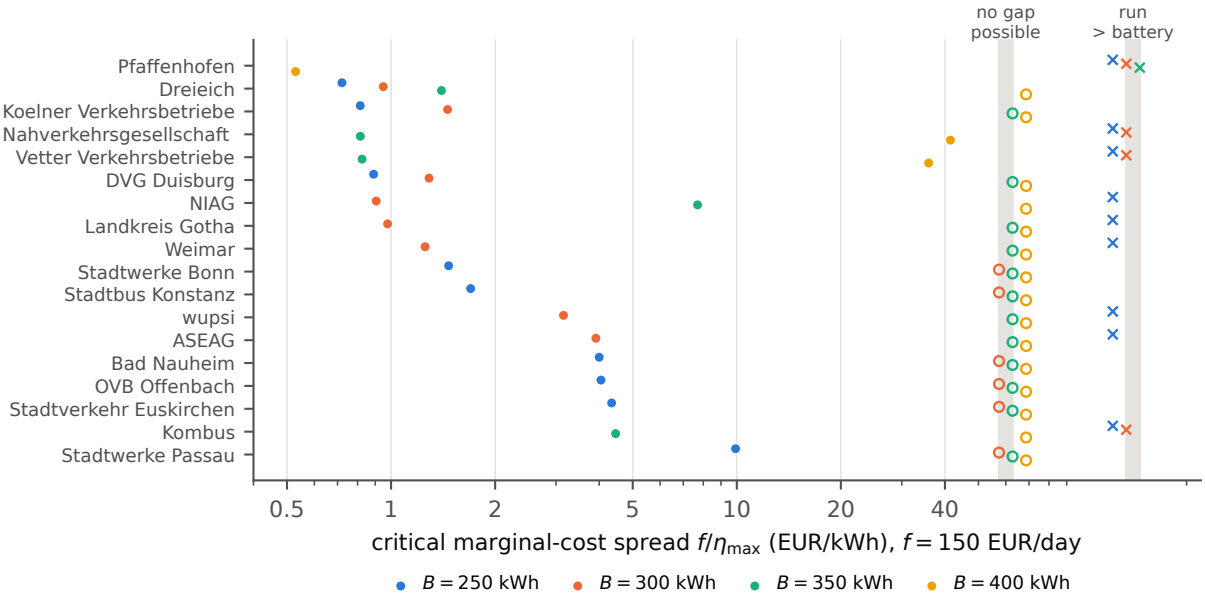


Figure 2: Critical marginal-cost spread $f/\eta_{\max}$ below which no support gap can occur (Proposition 6), for a vehicle cost of 150 EUR/day, per operator and usable battery size B. Hollow markers: $r_m = 0$ for all m, so no gap is possible at any price. Crosses: some peak run needs more than B, so the combination is outside the model.

1.40). At $B = 300$ kWh, 8 of 14 do, from 0.91 EUR/kWh (median 1.27). At $B = 250$ kWh, all 9 eligible operators do, from 0.72 EUR/kWh (median 1.70). These thresholds are several times the 0.15 EUR/kWh reference energy price. The thresholds describe the marginal spreads a proposed supply and fee model must reach at a feasible load. Medians here are conditional on operators with a positive matching increment at the specified capacity.

The offset grid and the exact failure intervals. At $B = 400$ kWh the grid has $18 \times 37 \times 4 = 2664$ scenarios. Only 7 have $\Delta > 0$, all with $s_0 > 0.3$ EUR/kWh, and the largest gap is 0.023% of total daily cost. At $B = 300$ kWh (14 eligible operators, 2072 scenarios) there are 6, with gaps up to 0.054%. In gap scenarios, the planner optimum's regret at its own marginal prices is a median 8.5 times the gap at $B = 400$ (4.2 at $B = 300$). The seven positive cells at 400 kWh all belong to Pfaffenhofen. Their modeled daily planning gaps range from 0.04 to 1.04 EUR after rounding. The operator's corresponding saving from deviating at the physical plan's posted marginal prices ranges from 0.30 to 4.56 EUR. These figures use the assumed supply curve and bus fee. Halving the midday budget leaves the reported gap counts unchanged.

The price grid can skip a narrow failure interval entirely. We evaluated Proposition 7 directly on the committed decimal run energies with rational arithmetic. For each eligible operator and each of the four slopes we intersected the open intervals with $[-0.2, 1.6]$ and divided their total length by 1.8. The last column of Table 1 averages that fraction equally over operator–slope combinations. It measures coverage of the specified offset range. Estimating frequencies in service would also require observations of the tariffs and marginal-cost curves depots face.

At 300 kWh the exact intervals cover a positive length for 19 of 56 operator–slope combinations, while the grid detects a gap for five. At 400 kWh these counts are four and three of 72. Interval

Table 1: Exposure to a positive gap within the same price-offset range. The main grid has 37 offsets per operator and slope. Both midday capacities gave the same entries in this table.

Battery (kWh)	Eligible operators	Operators with a gap		Mean range covered (%)
		On grid	Anywhere in range	
300	14	2	6	0.292
400	18	1	1	0.339

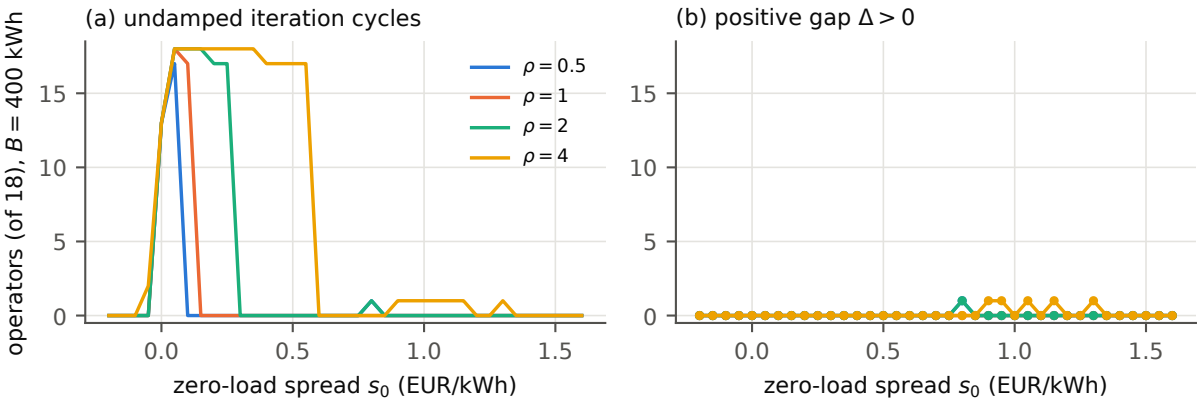


Figure 3: Number of operators (of 18; $B = 400\,\text{kWh}$, $f = 150\,\text{EUR/day}$, ample midday budget) whose undamped price iteration cycles (a) and whose planning problem has a positive gap (b), as functions of the zero-load spread s_0 and the price impact ρ. Cycling concentrates at small spreads with no gap.

membership agreed with the numerical objective comparison in every one of the 9,472 cells across both capacities and battery sizes, using the study's 10^{-9} gap threshold. As a separate classical check, a general linear assignment solver selected both endpoints of every real pair without using the sorted matching formula. At half (rounded down) and maximum matching cardinality for every eligible operator and battery size, its 64 objective values agreed with the formula within $6 \cdot 10^{-12}\,\text{kWh}$.

Price cycles under an endpoint response. Undamped iteration, started from the spread at the smallest attainable early load, cycles in 397 of the 2664 scenarios at $B = 400\,\text{kWh}$ (15%). Of these, 390 have $\Delta = 0$ (Figure 3). Classifying the hull optimum explains them: it lies inside the flat face in 384 cycling scenarios, inside a sloping face in 7 (exactly the gap scenarios) and at a vertex in 6, where the undamped step is too large for part (c) of Theorem 10. Every scenario with a face optimum cycles. The flat-face cases concentrate at small spreads: the cheapest fleet has spare midday capacity, and the vertex response jumps between the two ends of its interval. With half the midday budget, 189 scenarios (7%) cycle, 182 of them without a gap. At $B = 300\,\text{kWh}$ the pattern is the same: 307 of 2072 cycle and 301 of these have no gap. A study that counted price cycles as evidence of assignment nonconvexity would overstate it by a factor of about fifty here.

8 Duty commitment on a German distribution feeder

Matched design. The primary design has 72 assignments: two German price days (17 January and 17 July 2024), two depot sites, three charger sizes, two matched fleet classes, and three supply rules. The heterogeneous fleet resamples published Euskirchen peak-run energies by fixed midpoint ranks to 5–51 services per batch. The homogeneous control uses the same batch means. Both have 320 kWh usable batteries, efficiency 0.95, 125 kW plugs, a common one-hour midday window and a two-hour terminal window. These are constructed depot-scale schedules with complete successor compatibility.

We use the native radial SimBench rural medium-voltage feeder [23]. Each 15-minute snapshot retains native load and renewable profiles, taps, switches, shunts, voltage limits and terminal current ratings. SimBench's 2016 same-week/weekday profiles proxy the two 2024 operating dates; Energinet records are explicitly filtered to the DE price area and billed at their German delivery hour. A branch-flow conic relaxation [13] supplies numerical lower estimates; independent pandapower Newton flows check constructed whole-bus upper costs. No exactness theorem is presumed for this feeder.

Primary supply has hard native limits and no substitute supply. Two sensitivities allow a 1 MW unit at the depot at 414.7235 or 510.5304 EUR/MWh, linear fuel-cost proxies from a published 1 MW diesel unit and Germany's 2024 average diesel price [1, 8]. These are scenarios, not local bids or valuations of unserved service. A single constant depot MW per window is realized by plug-sharing with variable power, pausing and migration. We compare a predetermined rank pairing, an energy-aware minimum-bus matching chosen before prices, joint physical scheduling and convexification.

Cost brackets and numerical limits. The primary comparison resolves 66/72 assignments within one cent: the minimum-bus matching admits a whole-fleet schedule whose independently computed AC cost matches the numerical hull lower estimate. Four conic hulls are reported infeasible; two retain numerical bound widths of about 0.70 and 0.81 EUR. These are tolerance-based computations, with unverified exact-arithmetic dual feasibility. The first conic integer-count selection passed all strict checks in only three assignments. Its original statuses remain intact. Physical schedule construction and a second conic backend provide the separate brackets reported here.

The secondary one-hour terminal comparison resolves all 30 assignments with a numerical hull point; six are solver-infeasible. The additional 36 one-hour heterogeneous assignments fail a per-bus terminal charging condition. They are a design limitation and are excluded from the primary headline. Earlier residual failures are retained. Before date runs we replaced an unstable pure relative cone screen by a declared absolute-plus-relative screen in physical units, with independent AC agreement and native limits. Five repaired Gurobi primals still fail that cone screen. A relaxation can nevertheless bound a separately constructed physical point; solver status alone is never used as a physical witness.

Binding limits without extra buses. At the largest far depot, the hard-supply hull is solver-infeasible for both fleet classes on both days. With local backup the same 51-bus duties are AC-feasible and match the hull lower estimates within one cent. All twelve quarters reach the depot's native 0.965 p.u. floor; no native thermal constraint binds. Backup energy is 2.471 MWh in January and 2.241 MWh in July. Both window marginal prices equal the active backup offer within numerical tolerance. The absolute price rises, but the spread needed to buy an extra bus does not. The full wholesale ranges, 71.05 and 177.15 EUR/MWh, are also below the original homogeneous fixture's 527.78 EUR/MWh threshold. In the 36 distinct feasible wholesale comparisons, all choose

minimum fleet size. A predetermined pairing costs 0.40–6.54 EUR more in six January designs; the strong classical matching captures this benefit before prices are observed.

Actual price updates. We freeze and run price → fleet response → fixed-load OPF → price on the original 73 admitted hull assignments, including 51 primary assignments. All fleet responses use the complete count projection. Four rules start at the wholesale window prices: steps 1, 0.5 and 0.1 and $1/(k+1)$, with at most 100 iterations. Every grid price requires the physical conic screen and independent AC agreement. For steps 1, 0.5 and the diminishing rule, 43/73 reach a numerical fixed point; eight leave grid feasibility and 22 fail grid admission. Primary counts are 27/51 fixed, eight infeasible responses and 16 unresolved grid checks. With step 0.1, 16/51 reach the fixed-point test, eleven exhaust the iteration budget, and the same eight and sixteen fail feasibility and admission. A small damped price change alone is insufficient. Slow contracting sequences triggered an overly permissive approximate-recurrence detector for step 0.1; we retain those histories, withdraw their cycle labels and complete available traces using only cached grid responses. No nontrivial cycle is established. Prespecified flags cover positive support gap, price ambiguity, fleet ties and a verified step-size mechanism. Unresolved paths are not assigned a cause. Local two-sided price probes are recorded, with infeasible sides and solver failures retained. These results give no empirical support for the hypothesis of frequent cycles among the admitted fixed points.

Annual capacity and connection stakes. E-regio Netz serves Euskirchen. Its 2026 medium-voltage annual capacity rates are 12.22 and 88.64 EUR/kW-year below and above 2,500 utilization hours, respectively; its one-time connection contribution is 95.64 EUR/kW [11, 12]. The 53.63–549.04 kWh spread in mandatory early charging across pairings is not automatically a reduction in the depot's peak. For window lengths one and two hours, minimum peak is $\min_{E\in[\ell,U]} \max\{E, (S-E)/2\}$ in kW when energies are kWh. Every feasible matching at minimum fleet size admits the balanced point $E = S/3$ in all six primary heterogeneous designs and their homogeneous controls. Their minimum peaks are identical, ranging from 453 to 4,660 kW. The demonstrated annual capacity and connection savings from choosing between these covers are therefore zero in this fleet projection. Feeder feasibility and backup requirements are separate from that minimum-power calculation. Repeating these three-hour windows every day gives at most 1,095 utilization hours. Other depot loads, annual peaks and contracted connection capacity would be needed to value a different design.

9 Discussion

A decision at the depot. Given a proposed bus fee and a specified supplier cost curve, a planner can calculate the interval endpoints before choosing a coordination rule. Outside the open failure bands, a feasible fleet attains the hull optimum. At a supported tie the supplier can specify the desired quantity in addition to the price. Inside a band, let x_D be a physical optimum. At its own marginal price the fleet can save at least Δ by changing schedules. At a hull price p^*, the supplier can pay the operator $r(x_D; p^*)$ conditional on operating x_D. The payment is bounded above by Δ (Proposition 2). This compares two specified payment rules. Contract enforcement costs, participation rights and observed tariff parameters are needed to value a change in practice.

The new feeder study supports representative matching within its two-window class, while the hourly boundary test prevents a general recommendation to fix arbitrary zero-price duties.

The study illustrates why a ratio alone is insufficient for that decision. At 400 kWh, regrets in the seven positive grid cells are several times their planning gaps, yet the largest modeled saving

from deviating is 4.56 EUR per day. At 300 kWh, counting affected operators on the grid misses four operators with narrow failure bands. The interval calculation identifies both the exposed operators and the range of offsets to check.

For price-coordination algorithms. Theorem 10 separates two causes of non-convergence. On a flat face a supplier that knows the target load can request it at the supporting spread. A tie-breaking rule applied only at that spread cannot alter a trajectory that never reaches it. An undamped cycle may also surround a supported vertex. Under diminishing steps prices converge, and for an affine spread the step-weighted loads and fees recover the hull point. A supplier can distinguish these cases by checking whether a feasible schedule attains the target load *and* the hull vehicle cost.

Limitations. The exact theorem assumes two synchronized batches, common availability and efficiencies, zero-time transitions, ideal switching and period-aggregate settlement. Section 6 shows that edge-dependent transitions and a third batch are genuine boundaries, and that setup time requires separate treatment. The study derives run energies from public timetables and assumed energy rates. Estimating savings for a depot would require observed consumption, duty options and settlement terms. The 37-service certificate establishes existence beyond the class for one timetable with synthetic economics.

Open directions. Further work beyond the present tests could examine: a characterization for staggered charging windows, where serialization is no longer automatic, a measured application of the setup-time bound, and an analysis of a strategic fleet operator who anticipates the price change.

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A Proofs for Section 4

A.1 Proof of Theorem 3

Fix one allowed matching M with $|M| = m$, fleet $N = N_m$, requirement $R = \sum_{(i,j)\in M}(a_i + b_j - B)_+$ and $T = S_0 - NB$. Let x_v be a bus's net early battery gain. A paired bus serving A_i and B_j holds $B - a_i$ after its morning service; it cannot exceed capacity, so $x_v \le a_i$, and it must hold at least b_j before the afternoon service, so $x_v \ge a_i + b_j - B$. An A-only bus has $x_v \in [a_i - B, a_i]$, and a B-only bus, full before the early window, has $x_v \in [b_j - B, 0]$. These boxes are nonempty because $a_i, b_j \le B$. Their lower endpoints sum to T, their positive lower endpoints to R, and their upper endpoints to P; every upper endpoint is nonnegative.

Necessity. Whatever the order of actions, $x_v = \alpha C_v - X_v/\delta$ for the bus's early purchases C_v and exports X_v, so the total Q lies in $[T, P]$. A positive lower endpoint needs at least its value divided by α in purchases, hence $C \ge R/\alpha$. Exports are nonnegative, the shared connector carries gross energy $C + X \le K$, and full terminal replenishment needs the terminal purchase $L = (S_0 - Q)/\alpha \le H$. Terminal purchases are nonnegative because $Q \le P < S_0$.

Sufficiency. Fix totals (C, X) satisfying these inequalities with R in place of r_m. For any $Q \in [T, P]$ the boxes admit gains whose positive parts sum to $A = \max\{R, Q\}$. If $Q \ge R$, start from the positive lower endpoints and zero elsewhere, and distribute $Q - R$ through the remaining

headroom. If $Q < R$, keep the mandatory positive gains and distribute the negative total $Q - R$ through the negative lower intervals, whose combined lower endpoint is $T - R$. One-direction exchanges then realize $C_0 = A/\alpha$ and $X_0 = \delta(A - Q)$. Since $X \geq 0$ implies $\alpha C \geq Q$, and $\alpha C \geq R$, the residual $z = C - C_0$ is nonnegative and $X - X_0 = \alpha\delta z$.

Realize these residuals by battery-neutral cycles on one bus before its one-direction exchange. At early state of charge $s \in [0, B]$, use charge-then-discharge cycles of depth at most $B - s$ if $s \leq B/2$, and discharge-then-charge cycles of depth at most s otherwise. The permitted depth is at least $B/2 > 0$. Split the residual battery throughput αz into finitely many such chunks. Each returns to s, stays in $[0, B]$, and uses grid purchases depth$/\alpha$ and exports $\delta\cdot$depth, so together they realize exactly $(z, \alpha\delta z)$. Execute the cycles and then all one-direction exchanges serially: their gross energy is $C + X \leq K$, so they fit the common early window at the connector's power. The endpoint boxes guarantee the afternoon service, and nonnegative service draw preserves every intermediate state-of-charge bound. Refill serially at the terminal connector using total $L \leq H$. If $K = 0$ all early actions vanish; if $H = 0$, then $P < S_0$ makes every polygon empty, consistent with infeasibility.

Replacement. At fixed m, the quantities S_0, P, N_m, T and the fleet cost do not depend on the matching; only R does, and only through $C \geq R/\alpha$. Hence the polygon of a matching attaining r_m contains the polygon of every other matching of size m, and every point of it has just been realized. The union over matchings of size m therefore equals the polygon W_m, and the union over m gives $\mathcal{P}$. Each W_m is defined by five half-planes, so it has at most five vertices, and $\operatorname{conv}\mathcal{P}$ is the convex hull of their images. With exports prohibited, the construction above uses no exports when $X = 0$: then $\alpha C = Q \geq R$, so $A = Q$, $X_0 = 0$ and $z = 0$. □

Idle buses. Identical paid idle buses under the total fleet cap $n_A + n_B$ do not change $\mathcal{P}$. An idle bus has the box $[-B, 0]$. A plan with m pairs and q idle buses has $q \leq m$ by the cap; deleting q pairs and serving their services separately preserves N, S_0, P, T and the fleet cost while lowering R, and reproduces the same (C, X).

A.2 The lossless cases

Charge-only. With $\alpha = \delta = 1$ and $X = 0$, W_m becomes $\{C : \ C \geq r_m,\ C \leq K,\ S_0 - N_m B \leq C \leq P,\ C \geq S_0 - H\}$. Since $S_0 - N_m B$ is the sum of all lower endpoints and r_m the sum of their positive parts, $S_0 - N_m B \leq r_m$, and the layer is $\{E : \max\{\mathrm{Lo}, r_m\} \leq E \leq U\}$ with $L = S_0 - E$, which is (3). Feasibility is $\mathrm{Lo} \leq U$ because the empty matching has $r_0 = 0$.

Bidirectional. With $\alpha = \delta = 1$ and exports allowed, $E = C - X$ and $L = S_0 - E$. A point E is attainable in layer m if and only if $\max\{S_0 - N_m B, S_0 - H\} \leq E \leq P$ and some $C \geq \max\{r_m, E\}$ satisfies $2C - E \leq K$, that is, $E \leq K$ and $E \geq 2r_m - K$. The layer is therefore $[t_m, U]$ with $t_m = \max\{S_0 - H,\ S_0 - N_m B,\ 2r_m - K\}$ and $U = \min\{P, K\}$.

A.3 Monotonicity, convexity and the frontier

Deleting an edge from a minimum matching of size m shows $r_{m-1} \leq r_m$. For discrete convexity, consider the unit-capacity source–A–B–sink network on the allowed edges with costs $(a_i + b_j - B)_+$. Its minimum-cost flow LP is integral at integer flow values $m = 0, \ldots, \nu$, and its value is convex in the required flow, so $2r_m \leq r_{m-1} + r_{m+1}$. This uses bipartiteness, not completeness of Γ.

All thresholds at or below Lo are dominated in cost by m_0. The first retained increment is $r_{m_0+1} - \mathrm{Lo} \leq r_{m_0+1} - r_{m_0}$, and later increments are the original nondecreasing matching increments, all positive after m_0. The slopes $-f/\eta_m$ are therefore nondecreasing, followed by the flat segment. Endpoints are physical, interpolants are hull mixtures, and every physical branch lies on or above

this convex graph, so the graph is exactly the lower hull boundary. Consecutive breakpoints differ by one cardinality. For the bidirectional thresholds t_m, each term in the maximum defining t_m has a nondecreasing convex piecewise-linear extension in m. Their pointwise maximum is convex and nondecreasing. Its values at integer m therefore have nondecreasing increments, giving the same frontier construction starting at the last cardinality with $t_m = t_0$.

A.4 Proof of Corollary 5

The convex frontier plus strictly convex continuous G has a unique minimizer E^*. The empty matching admits every $E \in [\mathrm{Lo}, U]$. At Lo, at a breakpoint or on the flat segment, a physical plan attains the frontier, so $D = \mathrm{CH}$. Strictly between consecutive breakpoints the frontier's operating cost is strictly below that of every physical branch because $f > 0$, so no physical plan attains the hull optimum; compactness of the finite union of physical segments then gives $D > \mathrm{CH}$.

A.5 Sorted matching for the complete graph

Replacing a selected service by an unused smaller draw while keeping its partner cannot increase the nondecreasing edge cost, so some optimal size-m matching uses the m smallest draws in each batch. With $h(t) = (t - B)_+$ and $a \le a'$, $b \le b'$, convexity gives $h(a+b) + h(a'+b') \ge h(a+b') + h(a'+b)$. Swapping any two partners ordered in the same direction therefore cannot raise the cost, and repeated exchanges yield opposite order. For a non-complete graph this shortcut fails: with A draws $(5, 15)$, B draws $(14, 15)$ and $B = 20$, allowing only the index-matched edges gives $r_2 = 10$ instead of 9.

A.6 The worked example

The requirement matrix is $\left(\begin{smallmatrix} 0 & 0 \\ 9 & 10 \end{smallmatrix}\right)$, so $r_0 = r_1 = 0$, $r_2 = 9$, $S_0 = 49$, $\mathrm{Lo} = 0$ and $U = 15$. With $G(E) = 2401/20 - 9E/10 + E^2/10$, the physical minima of layers $m = 0, 1$ are at $E = 9/2$, with values $5841/40$ and $5561/40$; layer $m = 2$ is minimized at its endpoint $E = 9$ with value $2681/20$. On the sloping hull face the objective is $2821/20 - 151E/90 + E^2/10$, minimized at $E^* = 151/18$ with value $434201/3240$, so $\Delta = 121/3240$. The hull point mixes the three-bus plan at $E = 0$ and the two-bus plan at $E = 9$ with weights $11/162$ and $151/162$. At the physical optimum's marginal prices $(49/10, 4)$ each layer's bill is minimized at its lowest endpoint; bill plus vehicle cost is 224, 217 and $2181/10$, so the regret is $11/10$. Both mixed plans have explicit serial witnesses: at $E = 0$ pair the 5-draw A with the 14-draw B and refill $19, 15, 15$ on the terminal connector; at $E = 9$ the 15-draw A bus charges 9 at midday in the crossed pairing and both buses refill 20.

B Proof of Proposition 2

For any hull point (e, c), linear minimization gives $c + p^\top e \ge V(p)$. If $p \in \partial F(e(x))$, then $F(e) \ge F(e(x)) + p^\top(e - e(x))$. Adding and minimizing over the hull gives $\mathrm{CH} \ge V(p) + F(e(x)) - p^\top e(x) = c(x) + F(e(x)) - r(x; p)$, which is the regret bound; if $r(x; p) = 0$, then $D = \mathrm{CH}$.

Conversely let $D = \mathrm{CH}$. Compactness gives a physical minimizer x. The relative-interior condition gives strong Fenchel duality with an attained maximizer, $\mathrm{CH} = V(p^*) - F^*(p^*)$. Then

$$\underbrace{c(x) + (p^*)^\top e(x) - V(p^*)}_{\ge 0} + \underbrace{F(e(x)) - (p^*)^\top e(x) + F^*(p^*)}_{\ge 0} = D - \mathrm{CH} = 0,$$

so both terms vanish: Fenchel equality gives $p^* \in \partial F(e(x))$, and the first term gives $r(x; p^*) = 0$. The same identity holds for any physical optimum and any dual maximizer when $D > \mathrm{CH}$, with right-hand side Δ; since both terms are nonnegative, $r(x; p^*) \le \Delta$. □

C Proofs for Section 5

Throughout, $\bar{c}$ is the lower hull boundary on $[\mathrm{Lo}, U]$, extended by $+\infty$ outside, and $H(\pi) = \arg\min_E\{\bar{c}(E) + \pi E\}$ is the hull response. The finite candidate set $\mathcal{V}$ consists of all layer endpoints (ℓ_m, fN_m) and (U, fN_m). Since a linear function attains its minimum over $\operatorname{conv}\mathcal{P}$ at a point of $\mathcal{V}$, every $E \in R(\pi)$ satisfies $\bar{c}(E) = fN_m$ for its layer and $E \in H(\pi)$. The layer endpoints that lie on $\bar{c}$ are exactly the breakpoints Lo, ℓ_m $(m_0 < m \le m_{\max})$ and the point U, using the corresponding thresholds in the bidirectional case. The hull optimum satisfies $-\pi^* \in \partial\bar{c}(E^*)$ because G is differentiable.

C.1 Proof of Proposition 6

(i) By Corollary 5, $\Delta > 0$ places E^* strictly between consecutive breakpoints $u < u'$, where $\bar{c}$ is affine with slope $-f/(u'-u)$ because consecutive breakpoints differ by one cardinality. Optimality at an interior point gives $g(E^*) = f/(u'-u) \ge f/\eta_{\max}$. Strict increase of g and $E^* < u' \le U$ yield $g(U) > g(E^*)$.

(ii) On $[u, u']$ let $\phi = \bar{c} + G$. Then $\phi'(E) = -f/(u'-u) + g(E)$ is κ-Lipschitz and vanishes at E^*, so $\phi(E) - \phi(E^*) \le \kappa(E - E^*)^2/2$. Both u and u' are physical with cost $\bar{c}$, so $D \le \min\{\phi(u), \phi(u')\}$ while $\mathrm{CH} = \phi(E^*)$. One of u, u' is within $(u'-u)/2$ of E^*, which gives $\Delta \le \kappa(u'-u)^2/8$. The bound using $\eta_{\max}$ also holds when $\Delta = 0$. □

C.2 Proof of Proposition 7

Write $\tau_m = f/\eta_m$. On the open segment (ℓ_{m-1}, ℓ_m) the derivative of $\bar{c} + G_s$ is $s + h(E) - \tau_m$. It vanishes in the segment exactly for the intercepts in the corresponding open interval of (4). Strict convexity makes this stationary point the unique global optimum. Corollary 5 shows that these are precisely the positive-gap cases. As m increases the increments η_m are nondecreasing, including the first increment above Lo. Hence $\tau_{m+1} - h(\ell_m) \le \tau_m - h(\ell_m)$. The upper endpoint of band $m+1$ is at or below the lower endpoint of band m. Equality gives a shared boundary intercept whose optimal load ℓ_m is a physical breakpoint. Summing the lengths of the disjoint bands gives

$$\sum_{m=m_0+1}^{m_{\max}} \left[h(\ell_m) - h(\ell_{m-1})\right] = h(\ell_{m_{\max}}) - h(\mathrm{Lo}),$$

since $\ell_{m_0} = \mathrm{Lo}$. When $m_0 = m_{\max}$ both sides vanish. □

C.3 Proof of Lemma 9

Let $E \in R(\pi) \subseteq H(\pi)$, so $-\pi \in \partial\bar{c}(E)$, and recall $-\pi^* \in \partial\bar{c}(E^*)$. Monotonicity of the subdifferential of the convex function $\bar{c}$ gives $(-\pi + \pi^*)(E - E^*) \ge 0$, that is, $(\pi - \pi^*)(E - E^*) \le 0$. Since g is increasing and $g(E^*) = \pi^*$, $(\pi - \pi^*)(g(E) - \pi^*) \le 0$. For the second claim write $\pi_{k+1} - \pi^* = (1-a)(\pi_k - \pi^*) + a(g(E_k) - \pi^*)$. The two terms have opposite (weak) signs, and for real x, y with $xy \le 0$, $|x + y| \le \max\{|x|, |y|\}$. Since $E_k \in [\mathrm{Lo}, U]$, $|g(E_k) - \pi^*| \le M$. □

C.4 Proof of Theorem 10

We use three facts. (F1) The same monotonicity argument shows that $\pi < \pi'$, $E \in H(\pi)$ and $E' \in H(\pi')$ imply $E \geq E'$. (F2) If E^* lies in the relative interior of a maximal face $[w, w']$, then $\partial\bar{c}(E^*) = \{-\pi^*\}$, the face has slope $-\pi^*$, $[w, w'] \subseteq H(\pi^*)$, and $E^* \in H(\pi)$ only for $\pi = \pi^*$. By (F1), $H(\pi) \subseteq [\mathrm{Lo}, w]$ for $\pi > \pi^*$ and $H(\pi) \subseteq [w', U]$ for $\pi < \pi^*$. (F3) A fixed point $\hat{\pi} = g(\hat{E})$ with $\hat{E} \in R(\hat{\pi})$ has $-g(\hat{E}) \in \partial\bar{c}(\hat{E})$, so $\hat{E}$ minimizes the convex function $\bar{c} + G$; by uniqueness $\hat{E} = E^*$ and $\hat{\pi} = \pi^*$.

(a) By (F3), a fixed point requires $E^* \in R(\pi^*)$ to be a selectable layer endpoint on $\bar{c}$, that is, a breakpoint or U. Conversely, if E^* is a breakpoint or U, then $E^* \in H(\pi^*)$ is a candidate attaining the minimum, so $E^* \in R(\pi^*)$ and a selection choosing it makes π^* a fixed point. If $\Delta > 0$, every physical plan at E^* has a higher vehicle cost than $\bar{c}(E^*)$. If E^* lies in the open flat face $(\ell_{m_{\max}}, U)$, the cheapest layer attains the hull cost there but has no endpoint at E^*.

(b) Let σ be the selection, a function of π with values in $R(\pi)$; by (F1) it is nonincreasing, so $h = g \circ \sigma$ is nonincreasing and $h \circ h$ is nondecreasing. For $k \geq 1$, π_k lies in the finite set $g(\{E : (E, c) \in \mathcal{V}\})$. The odd and even subsequences are orbits of the nondecreasing map $h \circ h$, hence monotone, hence eventually constant in a finite set. Thus $\pi_{k+2} = \pi_k$ eventually. In the face case with $\pi_0 \neq \pi^*$, we show by induction that $\pi_k \neq \pi^*$: if $\pi_k \neq \pi^*$ then by (F2) $E_k \neq E^*$, so $\pi_{k+1} = g(E_k) \neq \pi^*$. By (F3) there is then no fixed point along the orbit, and the period is two. If $\pi_a > \pi^*$ is one state of the cycle, (F2) gives $E_a \leq w < E^*$, hence $\pi_b = g(E_a) < \pi^*$ and $E_b \geq w'$.

(c) By Lemma 9, $d_k = |\pi_k - \pi^*|$ satisfies $d_{k+1} \leq \max\{(1-a)d_k, aM\}$, so $d_k \leq \max\{(1-a)^k d_0, aM\}$ and the interval $d \leq aM$ is reached in finitely many steps and is invariant. Here $M > 0$ because $\mathrm{Lo} < U$ and g is strictly increasing. Under the step bound the interval contains no element of $\mathcal{T}$ distinct from π^*.

Vertex case. If $\pi^* \notin \mathcal{T}$, the unique minimizing load throughout the invariant interval is E^*, and each step multiplies the spread error by $1 - a$. If $\pi^* \in \mathcal{T}$, then E^* is one endpoint of the exposed face. Suppose E^* is its left endpoint and $v > E^*$ its right endpoint. In the interval above π^* the unique response is E^*, so the spread stays above π^* and converges geometrically. Below π^* the unique response is v, whose target $g(v) > \pi^*$ makes the spread reach or cross π^* after finitely many such steps. At equality, choosing E^* keeps the spread there and choosing any larger tied load sends it strictly above. Thus eventually every selected load equals E^*. The case with E^* the right endpoint follows by reversing the inequalities.

Face case: non-convergence. Suppose $\pi_k \to \bar{\pi}$. Then $a(g(E_k) - \pi_k) = \pi_{k+1} - \pi_k \to 0$, so $g(E_k) \to \bar{\pi}$. Since E_k ranges over a finite set and g is injective, $E_k = \bar{E}$ for all large k, with $\bar{\pi} = g(\bar{E})$. The set $\{\pi : \bar{E} \in R(\pi)\}$ is closed, being defined by finitely many weak linear inequalities, so $\bar{E} \in R(\bar{\pi})$, and (F3) gives $\bar{E} = E^*$ and $\bar{\pi} = \pi^*$. If E^* is not a breakpoint it is never selected, which is a contradiction. If it is a breakpoint, $E^* \in R(\pi_k)$ forces $\pi_k = \pi^*$ by (F2), so the orbit hits π^* exactly. For a fixed sequence of selected loads, π_k is affine in π_0 with slope $(1-a)^k \neq 0$, so each finite selection sequence admits at most one such π_0, and there are countably many sequences.

Face case, trapping. Inside the invariant interval, $\pi > \pi^*$ forces $E = w$ and $\pi < \pi^*$ forces $E = w'$. At $\pi = \pi^*$ every selected endpoint lies in $[w, w']$. With $d_- = \pi^* - g(w) > 0$ and $d_+ = g(w') - \pi^* > 0$, an update from $\pi > \pi^*$ gives $\pi' = (1-a)\pi + ag(w) \in [\pi^* - ad_-, \pi)$, with the symmetric inequality from below. A step from equality stays in $[\pi^* - ad_-, \pi^* + ad_+]$. This band is invariant. Repeated steps from one side approach a target strictly on the other side and hence enter the band in finite time. If the orbit eventually avoids the tie, or if ties always select w or w', the late responses belong to $\{w, w'\}$. An eventual constant load w would drive the spread to $g(w) < \pi^*$ and force the response w'. The symmetric argument excludes an eventual constant load w', proving recurrence of both.

(d) With $d_{k+1} \le \max\{(1-a_k)d_k, a_k M\}$, fix $\varepsilon > 0$ and k_ε with $a_k M < \varepsilon$ for $k \ge k_\varepsilon$. After k_ε, the interval $d \le \varepsilon$ is invariant, and outside it d contracts by the factors $1 - a_k$, whose product converges to zero because $\sum a_k = \infty$. Thus $\limsup d_k \le \varepsilon$ for every ε. Summing the update gives $\sum_{k<n} a_k(g(E_k) - \pi_k) = \pi_n - \pi_0$, which is bounded; dividing by $A_n = \sum_{k<n} a_k \to \infty$ and using $\pi_k \to \pi^*$ (Stolz–Cesàro) gives $\sum_{k<n} a_k g(E_k)/A_n \to \pi^*$. If g is affine, this average equals $g(\sum_{k<n} a_k E_k/A_n)$, and invertibility of g gives the load statement. Only finitely many layer endpoints exist. Each endpoint that is strictly suboptimal at π^* remains so in a neighbourhood of π^* by continuity. Thus every sufficiently late selected endpoint (E_k, c_k) satisfies $c_k + \pi^* E_k = \bar{c}(E^*) + \pi^* E^*$. Taking the same weighted average gives convergence of vehicle costs to $\bar{c}(E^*)$. □

Computational checks. The lossless charge-only implementation in `code/twobatch.py` accepts rational inputs. The tests in `code/test_twobatch.py` use exact arithmetic for matching, hull and gap calculations and for undamped cycles. They also check finite floating-point trajectories for damped and diminishing steps. The proofs above establish the conclusions for arbitrary run lengths and tie rules under their stated hypotheses.

D Proofs for Section 6

D.1 Common transition energy

Suppose every matched pair performs a compulsory task of energy $d \ge 0$, with nonnegative draw, during a noncharging interval after the early window and before the B batch. A paired bus then needs $x_v \ge a_i + b_j + d - B$ to finish its afternoon service, cannot exceed $x_v \le a_i$, and needs terminal refill $a_i + b_j + d - x_v$. The box is nonempty exactly when $b_j + d \le B$, so edges violating this are removed. Lower endpoints sum to $S_m - N_m B$ with $S_m = S_0 + md$, positive lower endpoints to the matching's requirement with d included, and the upper endpoints still to P. Because S_m depends only on m, the proof of Theorem 3 applies verbatim with S_0 replaced by S_m.

D.2 Proof of Proposition 11

For a retained edge, the paired bus has early gain interval $[a_i + b_j + d_{ij} - B, a_i]$. With m and z fixed, the lower endpoints sum to $S_0 + z - N_m B$, the upper endpoints to P, and the positive lower endpoints to $R(M)$. The serial realization in Appendix A therefore applies with $S_0 + z$ in the terminal balance as well. Only the purchase floor $C \ge R(M)/\alpha$ varies within the group, proving the replacement.

In the lossless charge-only case every point of a matching's branch satisfies $E + L = S_0 + z$. Nonempty branches with different totals are disjoint, which gives necessity. For the classical exchange equivalence, complete any two distinct rows and columns to two perfect matchings identical outside that 2×2 submatrix. Equality of their totals gives the exchange identity. Balancing each such exchange yields $d_{ij} = d_{i1} + d_{1j} - d_{11}$. Conversely, a perfect matching sums each row and column potential once, so its total is invariant. □

D.3 Proof of Proposition 12

A disjoint-block construction. Let $w_i = 2^i$ for $i = 0, \dots, k-1$, $W = \sum_i w_i$, $B = 20 \max_i w_i$. Block i has two A and two B services with draws $(B/2 + 3w_i,\ w_i)$ in each batch. Use windows A $[0, 1]$, early charging $[1, 2]$, noncharging transition tasks $[2, 3]$, B $[3, 4]$ and terminal charging $[4, 5]$; one early connector of budget $K = kB/2 + 4W$ and one terminal connector. Within a block, the identity

edges carry transition energy 0 and the cross edges $6w_i$. The fleet cap is $2k$, so exactly $2k$ buses are used and every bus serves one service from each batch. Identity requires early purchase $6w_i$ and consumes $B+8w_i$; cross requires none and consumes $B+20w_i$. For a set I of identity blocks, with $R = 6\sum_{i\in I} w_i$ and $S_{\max} = kB + 20W$, total consumption is $S_{\max} - 2R$, the per-bus early intervals sum to $[R, K]$, and the complete branch is $\{(E,\, S_{\max} - 2R - E,\, 2kf) : R \le E \le K\}$, each point realized by serial charging.

Complete graph. Now admit every A–B edge. Keep the within-block transition energies, and on each edge between different blocks set $d_{ij} = B - b_j$; raise the terminal budget to $H = 2kB$. For any edge put $e_{ij} = a_i + b_j + d_{ij}$ and $r_{ij} = (e_{ij} - B)_+$. Its early purchase interval is $[r_{ij}, a_i]$ whenever $b_j + d_{ij} \le B$, which holds for within-block edges because $B \ge 20 \max w_i$ and with equality between blocks. In the latter case $e_{ij} = a_i + B$ and $r_{ij} = a_i$: the bus refills completely at midday. For a perfect matching M these intervals sum to $[R_M, K]$ with $R_M = \sum_M r_{ij}$, total consumption is $S_M = \sum_M e_{ij}$, and the branch is $L = S_M - E$, $R_M \le E \le K$. Each terminal purchase is at most $e_{ij} - r_{ij} \le B$, so $H = 2kB$ accommodates it; $E = K$ is feasible for every perfect matching.

Give each high service (draw $B/2+3w_i$) in block i the potential $B/2+9w_i$ and each low service the potential w_i, as row potentials u_i and column potentials v_j; their total is $S_{\max}$. Every within-block edge satisfies $e_{ij} + 2r_{ij} = u_i + v_j$. A between-block edge has $e_{ij} + 2r_{ij} = B + 3a_i$, with slack over $u_i + v_j$ equal to $3B/2 - 9w_j$, $2B - w_j$, $B/2 + 2w_i - 9w_j$ or $B + 2w_i - w_j$ for the four high/low combinations. Each is at least $\gamma = w_{\max} + 2w_{\min} \ge 3$, the low-to-high case being limiting because $B/2 \ge 10w_{\max}$. Summing this assignment-potential certificate over M gives $S_M + 2R_M \ge S_{\max}$, with surplus at least γ if any between-block edge is used. Hence every physical point, and every mixture, satisfies $3E + L = S_M + 2E \ge S_M + 2R_M \ge S_{\max}$.

With $F(E, L) = 3E + L + (E-3)^2/(2K)$, total cost is at least $2kf + S_{\max}$. The equal mixture of the within-block corners $R = 0$ and $R = 6$ attains it at load $(3, S_{\max} - 9)$, so $\mathrm{CH} = S_{\max} + 2kf$. On a within-block branch the cost is $2kf + S_{\max} + 2(E - R) + (E-3)^2/(2K)$, increasing in E (derivative at least $2 - 3/K > 0$ since $K \ge 14$), hence minimized at $E = R$; the values $R = 6t$ nearest to 3 are 0 and 6, giving excess $9/(2K)$. Every between-block schedule has excess at least $\gamma \ge 3 > 9/(2K)$. Therefore $D = \mathrm{CH} + 9/(2K)$. □

D.4 Proof of Proposition 13

Reduce from numerical three-dimensional matching, strongly NP-complete [14, SP16]: three multisets of n positive integers must be partitioned into triples, one from each, each summing to B. We may assume total nB and every number at most B, mapping violations to a fixed NO instance. Make the multisets services in $[0, 1)$, $[1, 2)$ and $[2, 3)$ with those numbers as battery draws, with charging unavailable before time 3. Overlap and the fleet cap force exactly n buses, each serving one service per batch; a duty (a, b, c) is feasible exactly when $a + b + c \le B$, and the total nB forces equality. Conversely, a matching yields duties ending empty, which refill serially for B time units each during $[3, 3 + nB]$. The construction is polynomial and preserves strong hardness; membership in NP is immediate. □

E The certified public timetable

Sistig et al. [32] supply 37 mandatory Hildenbrand services and directed time and distance tables for five locations. We model service and deadhead traction at 1.2 and 0.96 kWh/km, respectively, plus 16 kWh/h during service, deadheading and off-depot waiting. Depot 15 alone permits charging and zero-draw parking. Up to 37 buses, each with 400 kWh usable energy, start full and return full by 30:00. They share one charge-only 360 kW connector with unit efficiency and zero switching

time. Each bus accepts 360 kW and costs 100 to use. Movement has zero monetary cost. The 30 hourly loads enter $F(e) = \sum_t (ae_t + be_t^2/2)$, where a and b are the exact stored binary64 values of 0.2 and $1/900$, respectively.

Upper bound on CH. In each model two feasible two-bus plans give a convex combination with objective at most 441.142238. For the original stored-energy model the weight minimizes the quadratic on the segment between the plans. For the additive model the checker evaluates that same fixed weight on two separately verified plans. Either feasible mixture suffices for the bound.

Bounds on D. One bus would have to serve all 37 trips in chronological order. None of the 36 consecutive connections offers a depot charge, and the services alone draw over 889 kWh. For three or more buses, full replenishment and nonnegative consumption imply a lower bound of three vehicle fees plus the linear cost of service energy, exceeding 477.82. For two buses, a 375-node tree branches on movement decisions. The checker reconstructs the rows and checks both children at each branch. Each leaf certifies infeasibility or uses a quadratic tangent, signed row multipliers and a residual bound over the variable bounds. These checks give $D \geq 441.532912$. A feasible two-bus plan gives $D \leq 441.532913$. With only the fee changed to $f = 100 + \delta$, $\delta \geq 0$, the mixture and the two-bus lower bound both rise by 2δ. The three-or-more-bus floor rises by 3δ, so the gap guarantee persists.

Scope. The original model uses the stored binary64 consumption of each service and leg on a finite graph of pull-out, direct, via-depot and pull-in modes with fixed leg times. The second model computes the same consumption formulas in exact rational arithmetic. A separate argument extends its two planning values to finite walks over the paired time–distance arcs with continuous waiting. The directed triangle inequalities and verified coverage of direct and via-depot connections allow a walk to be shortened while preserving its depot charging times. With common terminal access and zero-draw depot parking, charging can then be curtailed to respect capacity. This weakly reduces each hourly purchase and hence the monotone supply cost for each plan and each mixture. The certificate index in Appendix F identifies the separate checks of the source tables and charging access.

F Reproducibility

The feeder and duty-first sources, launchers, dependencies, repairs and archived outcomes are indexed in `paper/draft1/review/NOV3_REPRODUCIBILITY.md`. The record retains all denominators and the cached-only loop correction.

In `paper/draft1/`, `sh make.sh` rebuilds the sensitivity study, figures and paper from committed peak-run energies; `-tests` adds the original randomized tests. Setting `SISTIG_ROOT` reconstructs energies from Sistig et al. [32]. In `code/`, `twobatch.py` supplies the rational solver, `study.py` the sweep, and `gap_bands.py` checks every grid cell against the exact failure intervals. `check_review.py` reproduces the exceptional tie and the 64 general assignment checks.

For the 37-service certificates, `paper/focused-theory-20261029/CERTIFICATE_INDEX.md` gives the source revisions, trusted digests, export locations and offline commands. From an authorized copy of either export, run `verify.py` with CPython 3.12.2, the model's digest and a fresh output path as specified in that index. The author-run checks took 33 and 36 seconds with the standard library alone. The repository provides the index and receipts. Redistribution of the assembled exports and external replication remain open.